

Suggested Teaching Guidelines for
Microsoft .NET Technologies
PGCP-AC, August 2026

Duration: 120 Hours: 45 Theory (T) + 45 Lab (L) + 30 Self Learning (SL)

Objective: To acquire the knowledge of Microsoft.NET 9

Prerequisites: Knowledge of Computer Fundamentals.

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment – 20%

List of Course Material

Text book:

1. C#13 and .NET9, 2025 edition, Jefferson S. Motta / BPB

Reference book:

1. Programming ASP.NET Core, 1st edition, Dino Esposito / O'Reilly

Note: This module adopts C#13 and .NET9 as the programming standard for all sessions, lab work, and evaluations.

Session 1: .NET Framework Introduction (3T + 2SL)

- Introduction to the .Net Framework
- Intermediate Language (IL)
- Assemblies and their structure, EXEs/DLLs
- CLR and its functions
 - JIT Compilation
 - Memory Management and Garbage Collection
 - AppDomain Management
 - CLS, CTS
 - Security
- .Net Framework, .Net Core, Mono, Xamarin differences
- Managed and Unmanaged Code
- Introduction to Visual Studio
- Using ILDASM

Session 3: C# (2T + 4L + 2SL)

- C# Basics
- Project References, using
- Classes
- Data Types in .net and CTS equivalents
- Methods
 - Method Overloading
 - Optional Parameters
 - Named Parameters
 - Positional Parameters

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- Using params
- Local functions
- Properties
 - get, set
 - Readonly properties
 - Using property accessors to create Readonly property
- Constructors
- Object Initializer
- Destructors

Lab:

- Create a class that has Properties, Fields, Methods, Constructors (Trainer can specify any class of his choice, e.g. Student, Employee, etc)

Session 4: Inheritance, Interface and Static Members (3T + 4L + 2SL)

- Inheritance
 - Access Specifiers
 - Constructors in a hierarchy
 - Overloading in derived class
 - Hiding, using new
 - override
 - Sealed methods
 - Abstract Classes and Methods
 - Sealed Classes
- Static Members of a Class
 - Fields
 - Methods
 - Properties
 - Constructors
- Static Classes
- Static local functions
- Interfaces
 - Implementing an interface
 - Explicitly implementing an interface
 - Inheritance in interfaces
 - Default interface methods
- Operator overloading

Lab:

- Create multiple classes that use Inheritance based concepts
- Create and implement interfaces for the classes created in Lab 4
- Implement IDisposable, IComparable

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Session 6: Value Types, Reference Types and Arrays (2T + 2L)

- Reference and Value Types
- Value Types
 - struct
 - enum
- out and ref
- nullable types
- nullable reference types
- ?? and ??=
- Working with Arrays (single, multidim, jagged), Array Class members
- Indices and ranges
- Indexers

Lab:

- Lab based on array examples.
- Also create an array of the class created in Lab 1.

Session 7: Generics and Collection (3T + 3L + 2SL)

- Generic classes
- Generic methods
- Generic Constraints
- Collections – generic and non-generic
- Collection Examples based on ICollection, IList, IDictionary (both generic and non-generic)
- Iterating collections using foreach
- Using Tuples to pass multiple values to a function

Lab:

- Lab based on collection examples.
- Also create a collection of the class created in Lab 1.

Session 8: Delegates and Lambdas (2T + 2L + 2SL)

- Delegates
 - Calling methods using delegates
 - Uses of delegates
 - Multicast delegates
 - Action, Func, Predicate delegates
- Anonymous methods
- Lambdas

Lab:

- Lab based on delegates examples.

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Session 9: Exception Handling (2T + 2L + 2SL)

- Error Handling (Exceptions Handling)
 - Checked & Unchecked Statements
 - The try, catch, finally clauses
- User Defined Exception classes
- Declaring and raising events
- Handling events

Lab:

- Lab based on exceptions and events examples.

Session 10: LINQ (2T + 2L + 2SL)

- Anonymous types
- Extension methods
- Partial classes
- Partial methods
- LINQ to objects
- Writing LINQ queries
- Deferred execution
- LINQ methods
- PLINQ

Lab:

- Lab based on LINQ examples

Session 11: Assemblies, Reflection and File I/O (2T + 2SL)

- Creating a shared assembly
- Creating Custom Attributes
- Using Reflection to explore an Assembly
- Using Reflection to load an Assembly dynamically
- Files I/O and Streams
 - Working with drivers, Directories, and Files
 - Serialization and De-serialization
 - Custom serialization with ISerializable

Session 12: Multithreading (4T + 4L)

- Threading
 - ThreadStart, ParameterizedThreadStart
 - ThreadPool
 - Synchronizing critical data using lock, Monitor and Interlocked

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- Working with Tasks
 - Calling functions with and without return values
 - Using async, await
- Using the Task Parallel Library
- Calling functions with and without return values
- Task.Run and Task.Factory.StartNew
- Task Cancellation
- Continuation Tasks
- Exception Handling
- Using async, await
- Using the Task Parallel library
- Parallel.For, ForEach and Invoke

Lab:

- Threading related examples
- Task related examples

Session 14: Data Management with ADO.NET (2T + 2L + 2SL)

- Microsoft.Data.SqlClient introduction
- Connection, Command, DataReader, Transaction
- Asynchronous command Execution
- Asynchronous Connections
- DataAdapter, DataSet and DataTable

Lab:

- Lab based on ADO.NET

Session 15: ASP.NET MVC Core (2T + 4L + 2SL)

- Introduction to Asp.Net MVC CORE
 - Architecture of an ASP .Net MVC application
 - Understanding Folder structures and configuration files
 - Understanding Request life cycle
 - App Pipeline
 - Coventional Routing
 - Route data
 - Using Attribute Routing
- Understanding Controllers and Action
 - Create a controller
 - How actions are invoked
 - HttpGet , HttpPost , NoAction Attributes
 - Running Action result
 - Derived classes of ActionResult

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- HttpStatusCode
- Understanding Views
 - Creating Views
 - ViewStart
 - Layouts
- Understanding ViewBag
- Writing Razor codes in views

Lab:

- Lab based on creating a mvc application

Session 16: Models (2T + 2L)

- Creating Models & ViewModel
- Creating Strongly Typed Views
- Using Various Scaffold Templates
- Validation using Data Annotations
- Client side and server side validation
- CRUD operation using Model
- Partial View

Lab:

- Lab based on models

Session 17: MVC State Management (2T + 2L)

- ViewBag , ViewData, TempData
- Session
- Cookies
- Caching
- Configuring Sessions
- HttpContext

Lab:

- Lab based on concepts covered in the class

Sessions 18 & 19: Entity Framework (4T + 4L + 4SL)

- Introduction to EF
- Using DBFirst and Code First Approach
- CodeFirst using various Data Annotations
- Using Validation, Primary Key , Foreign Key etc
- CodeFirst using Fluent APIs
- Database Migrations
- CRUD operation using EF

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- Change tracking

Lab:

- Lab based on Entity Framework

Session 20: MVC Security (2T + 2L)

- Windows Authentication
- Cookie Authentication
- JWT Authentication
- Identity
- OAuth
- Using Authorize & Allow Anonymous attributes
- Role based authorization
- Policy based authorization
- Claims based authorization
- Preventing Forgery Attack using AntiForgeryToken
- Preventing Cross Site Scripting Attack
- Filters and Custom Action filter

Lab:

- Lab based on concepts covered in the class

Session 21: Web APIs (2T + 2L + 4SL)

- Localization in MVC (Demo Only)
- Deploying ASP.NET MVC application (Demo only)
- Web APIs
 - Creating ASP.NET MVC Web API
 - Configuring for CORS
 - Different Verbs
 - Consuming using a client
 - Using Newtonsoft APIs
 - Integrating Web API with React App
 - Configuring CORS of React App and Web API
 - Sending request from React App, processing at Web API and effecting the database

Lab:

- Create a RESTful service using Web API.
- Create a consumer.

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Session 22: MVC Integration with React (4T + 4L + 2SL)

- Introduction to MVC and React
- Setting up the Project
- Integrating React with MVC Backend
 - Define Models
 - Implement Controllers
 - Use Views
- Data Management and State Handling
 - Establish data flow
 - Manage state
- Advanced Topics and Best Practices
 - Authentication and Authorization
 - Routing
 - Structuring React components

Lab:

- Create a RESTful service using WEB API with React as the front end.